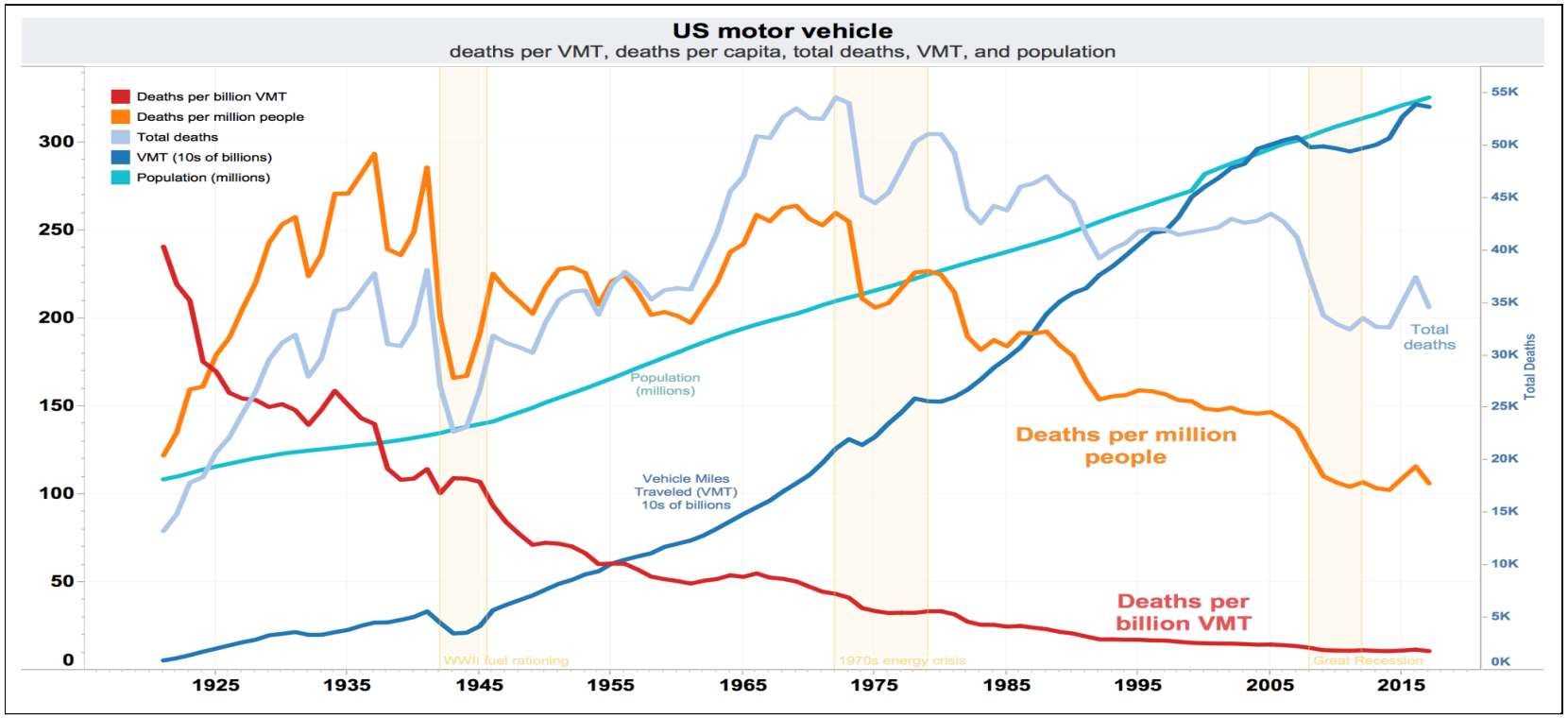
Driven
Hazen Capstone 2

- Problem statement: In considering the transformation of rural areas into nonrural areas in the US, what factors most strongly predict fatal crashes in each context for the year under consideration and thereby what is predictable as this transformation continues?

- Context: Since the 1970s, motor vehicle deaths have declined. In 2017 through 2019, the US saw steady declines in the per capita fatalities from vehicles, but in 2020 and 2021 the US saw an increase of 7.1% and 10.5% respectively (Wikipedia). Advanced safety measures such as air bags, ABS brakes, and electronic stability control have all helped reduce fatalities and injuries, but we need to search for other predictors of fatal crashes so that appropriate policies and safety measures can be enacted.



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- Criteria for success: If the ML model can predict fatalities given certain criteria, we can model how many fatalities will result from the continuing transformation of rural areas into more populated communities. Both a slide deck and a final report will result from this project.

- Scope of solution space: The data will come from just the US and primarily look at the differences between rural and nonrural areas in light of many other factors. Scope is limited by only having a binary divide in rurality although the reality is more complex. For this Capstone 2 project, a time-series analysis is not the goal. Considering the impact of the pandemic, the year 2019 was chosen as the last "normal" year where data is available.

- Constraints: Major contributors to crashes, such as technology use and distracted driving, are not part of the data. Such contributing factors may be similar in rural and nonrural areas.

- Stakeholders: NHTSA policy directors; State departments of transportation; Zoning councils in rural areas

- Data sources: The [National Center for Statistics and Analysis]

  Following the data science pipeline, this analysis will restrict itself to the accident table of the NHTSA FARS datasets from 2019 (FARS2019NationalCSV.zip): https://www.nhtsa.gov/file-downloads?p=nhtsa/downloads/FARS/2019/National/

  After munging the data to utf-8 format, the accident table was imported as a pandas DataFrame and has 33,487 rows across 91 columns. Initial data wrangling will reduce the number of influential factors, and EDA will trim them further. For this analysis, several types of linear regression will be applied to the data to find the best functioning model. In the end, both a slide deck and final report will be produced. Fatalities is one of the key columns for this DataFrame. In addition, at least initially, rural/urban,

drunk driving, weather, light condition, work zone, and pedestrian columns all appear to be important for this analysis. Some are as ‘object’ currently and some are as integers; properly adjusting each as either an integer or as a multi-level factor will be important for the analysis.

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